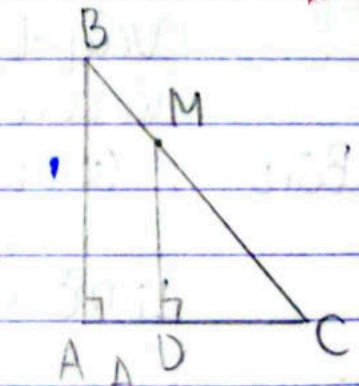
Ngọc Minh BTUN1

Câu 1: Xét $\triangle ABC$ và $\triangle MDC$ có:

$\widehat{BAC} = \widehat{MDC} (= 90°)$

$\widehat{ACB}$ chung

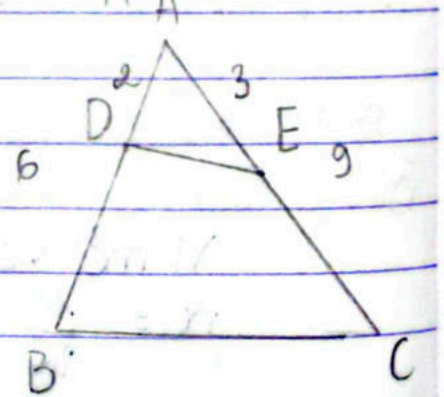
$\Rightarrow \triangle ABC \backsim \triangle MDC$ (gg)



Câu 2: Ta có: $\frac{AD}{AB} = \frac{2}{6} = \frac{1}{3}$

$\frac{AE}{AC} = \frac{3}{9} = \frac{1}{3}$

$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}$

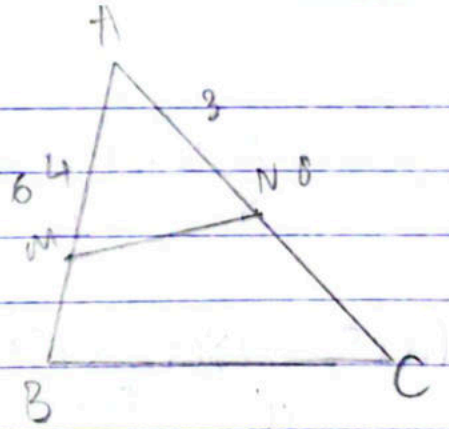


$\Rightarrow DE // BC$ (Định lý Thales đảo)

Được quét bằng CamScanner

Date . . . No . .

Câu 3:



Ta có: $\frac{AM}{AC} = \frac{4}{8} = \frac{1}{2}$

$\frac{AN}{AB} = \frac{3}{6} = \frac{1}{2}$

$\Rightarrow \frac{AM}{AC} = \frac{AN}{AB}$

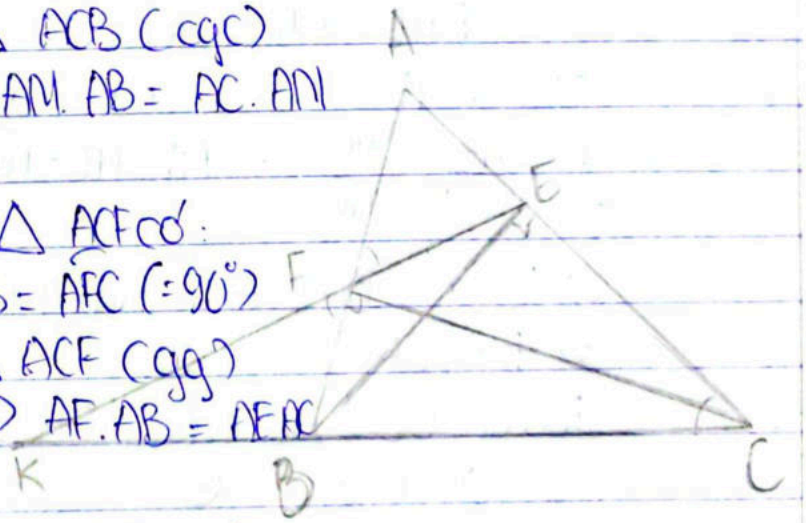
Xét $\triangle AMN$ và $\triangle ACB$ có:

$\frac{AM}{AC} = \frac{AN}{AB}$ (cmt), $\widehat{BAC}$ chung

$\Rightarrow \triangle AMN \backsim \triangle ACB$ (cgc)

$\Rightarrow \frac{AM}{AC} = \frac{AN}{AB} \Rightarrow AM.AB = AC.AN$

Câu 4:



a) Xét $\triangle ABE$ và $\triangle ACF$ có:

$\widehat{BAC}$ chung, $\widehat{AEB} = \widehat{AFC} (= 90^\circ)$

$\Rightarrow \triangle ABE \backsim \triangle ACF$ (gg)

$\Rightarrow \frac{AF}{AE} = \frac{AC}{AB} \Rightarrow AF.AB = AE.AC$

b) Vì $\triangle ABE \backsim \triangle ACF$ (cmt)

$\Rightarrow \frac{AE}{AB} = \frac{AF}{AC}$

Xét $\triangle AEF$ và $\triangle ABC$ có:

$\frac{AE}{AB} = \frac{AF}{AC}$, $\widehat{BAC}$ chung

$\Rightarrow \triangle AEF \backsim \triangle ABC$ (cgc)

$\Rightarrow \widehat{AEF} = \widehat{ABC}$ (2 góc t/ứ)

c) Vì $\triangle AEF \backsim \triangle ACB$ (cmt)

$\Rightarrow \widehat{AFE} = \widehat{ACB}$

Mà $\widehat{KFB} = \widehat{AFE}$ (đđ)

$\Rightarrow \widehat{ACB} = \widehat{KFB}$

Chay $\widehat{KCE}$)

Xét $\triangle KCE$ và $\triangle KFB$ có: $\widehat{EKC}$ chung, $\widehat{KCE} = \widehat{KFB}$

Được quét bằng CamScanner

Date . . . No . .

$\Rightarrow \triangle KCE \backsim \triangle KFB$ (gg)

$\Rightarrow \frac{KF}{KB} = \frac{KC}{KE} \Rightarrow KF.KE = KC.KB$

Câu 5: a) Xét $\triangle ADB$ và $\triangle AEC$

có: $\widehat{ADB} = \widehat{AEC} (=90°)$

$\widehat{BAC}$ chung

$\Rightarrow \triangle ADB \backsim \triangle AEC$ (gg)

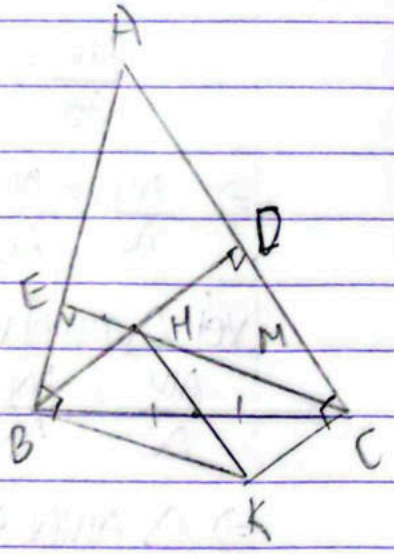
b) Xét $\triangle HEB$ và $\triangle HDC$

có: $\widehat{BEH} = \widehat{HDC} (=90°)$

$\widehat{BHE} = \widehat{CHD}$ (đđ)

$\Rightarrow \triangle HEB \backsim \triangle HDC$ (gg)

$\Rightarrow \frac{HE}{HD} = \frac{HB}{HC} \Rightarrow HE.HC = HD.HB$



c) Ta có: $BD \perp AC$ (hay $BH \perp AC$) và $CK \perp AC$

$\Rightarrow BH // CK$ ①

Ta có: $CE \perp AB$ (hay $CH \perp AB$) và $BK \perp AB$

$\Rightarrow CH // BK$ ②

Từ ①, ② suy ra BHCK là hbh

Mà M là tđ BC

$\Rightarrow M_{CK}$ là tđ HK

$\Rightarrow H, M, K$ th.

Được quét bằng CamScanner